

RReLU#

<https://pytorch.org/docs/stable/generated/torch.nn.RReLU.html#torch.nn.RReLU>

Description:

Applies the randomized leaky rectified linear unit function, element-wise. Method described in the paper: Empirical Evaluation of Rectified Activations in Convolutional Network. The function is defined as: where a is randomly sampled from uniform distribution $U(\text{lower}, \text{upper})$ during training while during evaluation a is fixed with $a = \frac{\text{lower} + \text{upper}}{2}$. lower (float) – lower bound of the uniform distribution. Default: $-\frac{1}{8}$

Function Signature

```
class torch.nn.RReLU(lower=0.125, upper=0.3333333333333333,  
inplace=False)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.RReLU(0.1, 0.3)  
>>> input = torch.randn(2)  
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the randomized leaky rectified linear unit function, element-wise.

Method described in the paper: Empirical Evaluation of Rectified Activations in Convolutional Network.

The function is defined as:

where a is randomly sampled from uniform distribution $U(\text{lower}, \text{upper})$ during training while during evaluation a is fixed with $a = \frac{\text{lower} + \text{upper}}{2}$.

lower (float) – lower bound of the uniform distribution. Default: $\frac{1}{8}$

upper (float) – upper bound of the uniform distribution. Default: $\frac{1}{3}$

inplace (bool) – can optionally do the operation in-place. Default: False

Input: $(*)^{(*)} (*)$, where $*$ means any number of dimensions.

Output: $(*)^{(*)} (*)$, same shape as the input.

Return the extra representation of the module.

Runs the forward pass.

